

Performance

Bandwidth

One characteristic that measures network performance is bandwidth.

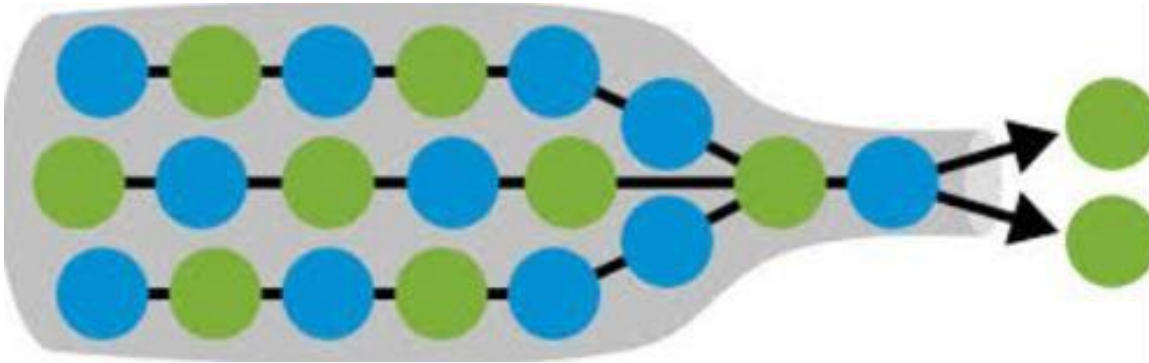
In networking, we use the term bandwidth in two contexts.

- ❑ The first, bandwidth in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- ❑ The second, bandwidth in bits per second, refers to the speed of bit transmission in a channel or link.

Throughput

The throughput is a measure of how fast we can actually send data through a network.

$$\textit{Throughput} = \frac{\text{Number of Units Produced}}{\text{Time Period}}$$



Throughput

- Although, bandwidth and throughput both in bits per second, they are different.
- The bandwidth is a potential measurement of a link; the throughput is an actual measurement of how fast we can send data.
- For example, we may have a link with a bandwidth of 1 Mbps, but the devices connected to the end of the link may handle only 200 kbps. This means that we cannot send more than 200 kbps through this link.

Throughput

Bandwidth	Throughput
Bandwidth refers to the maximum amount of data that can be transmitted at a given period of time.	Throughput is the actual amount of data that can be transmitted at a defined period of time.
Bandwidth mainly refers to a theoretical peak value.	Throughput refers to an achieved value.
It refers to the theoretical delivery of the packet data over a communication channel.	It confirms the actual delivery of the packet data over a communication channel.
It is usually measured in bits per second or bytes per second.	It is measured in any quantifiable unit related to the process

Throughput

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

$$\text{Throughput} = \frac{12,000}{60} \times \frac{10,000}{8} = 2 \text{ Mbps}$$

Latency (Delay)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source. We can say that latency is made of four components:

- propagation time,
- transmission time,
- queuing time, and
- processing delay.

Latency = propagation time + transmission time + queuing time + processing delay

Latency (Delay)

Propagation Time

Propagation time measures the time required for a bit to travel from the source to the destination. The propagation time is calculated by dividing the distance by the propagation speed.

$$\text{Propagation time} = \frac{\text{Distance}}{\text{Propagation speed}}$$

Latency (Delay)

Propagation Time

What is the propagation time if the distance between the two points is 12,000 km?
Assume the propagation speed to be 2.4×10^8 m/s in cable.

Solution

We can calculate the propagation time as

$$\text{Propagation time} = (12000 \times 1000) / (2.4 \times 10^8) = 50 \text{ ms}$$

The example shows that a bit can go over the Atlantic Ocean in only 50 ms if there is a direct cable between the source and the destination.

Latency (Delay)

Transmission Time

In data communications we don't send just 1 bit, we send a message. The first bit may take a time equal to the propagation time to reach its destination; the last bit also may take the same amount of time. However, there is a time between the first bit leaving the sender and the last bit arriving at the receiver. The first bit leaves earlier and arrives earlier; the last bit leaves later and arrives later. The time required for transmission of a message depends on the size of the message and the bandwidth of the channel.

$$\text{Transmission time} = \frac{\text{Message Size (bits)}}{\text{Bandwidth (bps)}}$$

Latency (Delay)

What are the propagation time and the transmission time for a 2.5-kbyte message (an email) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 ms.

Solution

We can calculate the propagation and transmission time as

$$\text{Propagation time} = (12000 \times 1000) / (2.4 \times 10^8) = 50$$

$$\text{ms Transmission time} = (2500 \times 8) / 10^9 = 0.020 \text{ ms}$$

Latency (Delay)

What are the propagation time and the transmission time for a 5-Mbyte message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Solution

We can calculate the propagation and transmission times as

$$\text{Propagation time} = \frac{12,000 \times 1000}{2.4 \times 10^8} = 50 \text{ ms}$$

$$\text{Transmission time} = \frac{5,000,000 \times 8}{10^6} = 40 \text{ s}$$

Latency (Delay)

Queuing Time

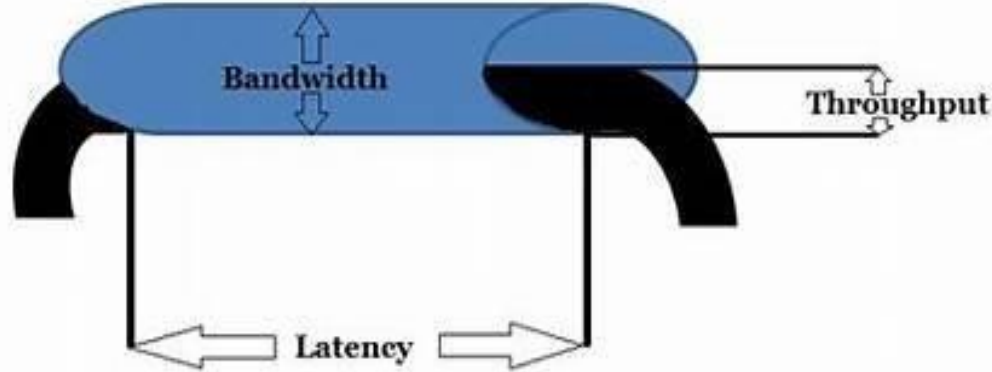
- The third component in latency is the queuing time, the time needed for each intermediate or end device to hold the message before it can be processed. The queuing time is not a fixed factor; it changes with the load imposed on the network. When there is heavy traffic on the network, the queuing time increases.
- An intermediate device, such as a router, queues the arrived messages and processes them one by one. If there are many messages, each message will have to wait.

Processing delay

- The time taken by network devices (such as routers and switches) to analyze the packet header, determine the route, and perform error checking.
- Example: If a router takes 2 ms to process a packet, this delay adds up to the total latency.

Latency (Delay)

- Network bandwidth refers to the capacity of the network for data to be moved at one time.
- Throughput expresses the amount of data.
- Latency refers to the speed at which data is transmitted.



Data Flow

Data flow is the direction of data

It can be

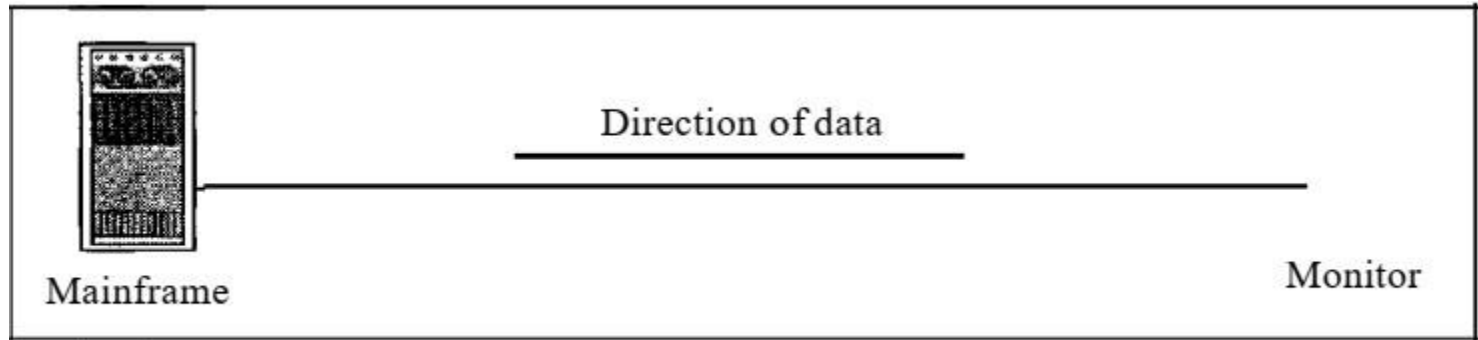
- Simplex
- Half duplex
- Full duplex

Data Flow

Simplex

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive.

Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.



a. Simplex

Data Flow

Half-Duplex

In half-duplex mode, each station can both transmit and receive, but not at the same time. :

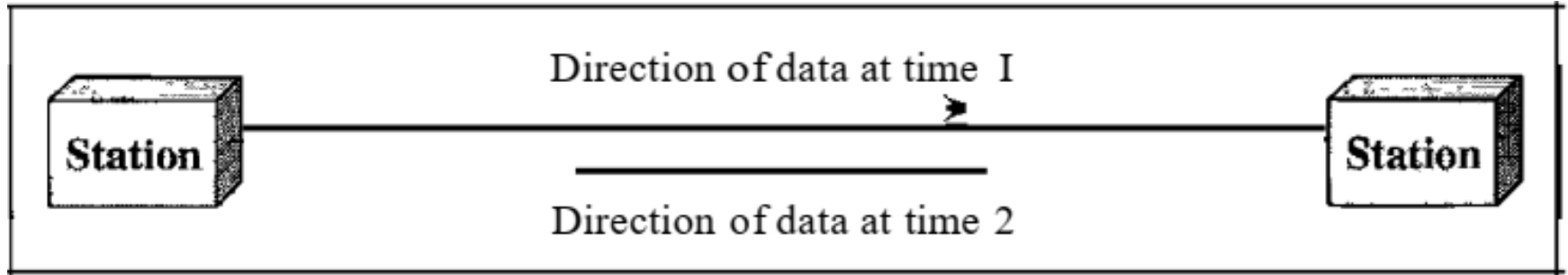
When one device is sending, the other can only receive, and vice versa.

The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait. In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

Data Flow

Half-Duplex



b. Half-duplex

Data Flow

Full-Duplex

In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously. The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.

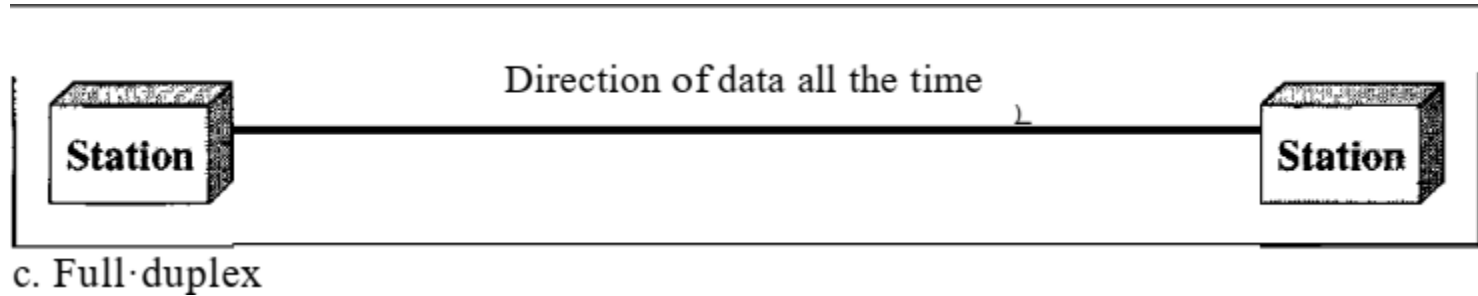
This sharing can occur in two ways:

Either the link must contain two physically separate transmission paths, one for sending and the other for receiving;
or the capacity of the channel is divided between signals traveling in both directions.

One common example of full-duplex communication is the telephone network. The full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.

Data Flow

Full-Duplex



Reference

Data Communication and Networking
-Behrouz A Forouzan